GOKULAHEALTH

V2 Product Requirements Document

Product Management Case Study | V1 → V2

Product Definition & Requirements

Product	GokulaHealth
Artifact	Product Requirements Document (PRD)
Phase	Product Definition
Version	V2
Status	Draft for Validation
Document Owner	Bhagyashree Reddy
Source	V1 Analysis + Initial User Research

1. Document Purpose

- **Context:** Defines the V2 direction for GokulaHealth based on V1 analysis, initial user research with one prospective participant, and product definition.
- **Objective:** Translates user problems, V1 gaps, and product definition opportunities into clear functional requirements, user stories, risk dependencies, and release approach.

Note: Serves as a draft for validation rather than a fully market-validated specification due to limited research data.

2. Product Context

- **App Overview:** Mobile application designed for cattle management, milk-production tracking, health records, and analytics.
- **V1 Capabilities:** Supports cattle registration, morning/evening milk production logging, milk diary records, vaccination details, individual trends, and factory-level comparisons.

V2 Strategy: Focuses on reducing record-keeping friction, providing clear monthly production summaries, improving analytics readability, and ensuring reliable offline operation for core workflows.

3. Problem Statement

Cattle/dairy operators may maintain milk-production records manually, which can make monthly calculation and analysis time-consuming.

- **Usability Challenge:** Current analytics and production experience can be difficult to interpret quickly for monthly performance reviews.

Validation Note: Automated monthly calculations may be useful but require broader testing with active operators.

4. Target User

4.1 Primary User

Small-scale cattle/dairy operators or cattle owners responsible for daily record-keeping and monitoring milk production.

4.2 Potential Secondary Users

- Family members involved in cattle management
- Farm workers
- Small dairy operators
- Larger/commercial cattle-farm operators

These are potential segments rather than fully validated segments. V2 requirements should be revisited if research shows that different user groups have substantially different workflows.

5. User Research Basis

- **Discovery Findings:** Initial discovery involved one prospective user planning a large-scale cattle milk business. The participant indicated that records would likely be maintained in a notebook, that monthly calculation and analysis would take time, and that automatic monthly calculation/analysis would be useful.
- **Key Priorities:** Monthly totals, monthly averages, and morning-vs-evening comparisons were identified as initial priority insights.

Limitations: Current insights are based on a single prospective participant, requiring further validation with active cattle/dairy operators.

6. V1 Findings Driving V2

V1 Finding	Product/User Impact	V2 Response
Milk diary already captures morning/evening production.	Strong alignment with the core record-keeping problem.	Improve entry flow and make resulting summaries easier to understand.
Monthly production information exists but can be difficult to understand quickly.	Reduces the value of captured data.	Make monthly total and average prominent and clear.
Analytics has readability/context issues.	Users may struggle to interpret production information.	Improve labels, contrast, hierarchy, and explanatory context.
30-day cattle trend is unclear.	Purpose and interpretation are uncertain.	Validate need before redesigning/expanding.
Factory comparison is incomplete/unclear.	User value is not established.	Keep lower priority until validated.

7. Product Goal

Make milk-production records easier to maintain and monthly production performance easier to understand.

8. V2 Product Objectives

- Reduce friction in daily milk-production recording.
- Make monthly total production easy to find and understand.
- Make monthly average production easy to find and understand.
- Provide a clear morning-versus-evening production view.
- Improve the readability and interpretation of production analytics.
- Create a foundation for later validated cattle-level insights.

9. Scope

9.1 In Scope — V2 MVP

- Milk-production entry for morning and evening yield.

- Clear saving/confirmation of production records.
- Monthly total production summary.
- Monthly average production summary.
- Morning-versus-evening production comparison.
- Improved production analytics presentation and context.
- Usability improvements to the milk-production workflow.
- Offline-first operation for the core production workflow.

9.2 Secondary / Validate Before Commitment

- Individual cattle 30-day production trend improvements.
- Cattle-level production comparison.
- Factory-level production comparison.
- Additional advanced analytics.

9.3 Out of Scope for This V2 Definition

- Cloud synchronization as a core V2 scope item.
- Multi-user account/role management.
- Payment or subscription functionality.
- Broad AI analytics without validated user need.
- Large-scale enterprise farm management features.

These items may be considered later, but they should not distract from validating the core milk-production problem.

10. Functional Requirements

ID	Requirement Area	Requirement
FR-01	Milk Production Entry	The user must be able to enter morning milk production for a cattle record/date.
FR-02	Evening Production Entry	The user must be able to enter evening milk production for a cattle record/date.
FR-03	Save Record	The user must be able to save a production record and receive clear confirmation.
FR-04	Record Review	The user must be able to review previously recorded production information.
FR-05	Monthly Total	The product must calculate and display total milk production for the selected month.
FR-06	Monthly Average	The product must calculate and display average milk production for the selected month.
FR-07	Morning vs Evening	The product must provide a clear comparison of morning and evening production.
FR-08	Analytics Clarity	Production analytics must use readable labels, sufficient contrast, and understandable context.
FR-09	Offline Core Workflow	Core production recording and review must remain usable without an internet connection.
FR-10	Data Integrity	Saved production values must remain consistent with the records entered by the user.

11. Non-Functional Requirements

- Usability: A user should be able to understand the primary production workflow without requiring technical knowledge.
- Readability: Text, values, labels, and charts must have sufficient visual contrast and clear hierarchy.
- Offline reliability: Core production workflows should function without network connectivity.
- Performance: Production screens and summaries should respond without noticeable delay during normal use.
- Data consistency: Calculated summaries should be derived from saved production records and remain consistent with the underlying data.
- Maintainability: V2 changes should fit the existing Android architecture where practical rather than introducing unnecessary technical complexity.

12. User Stories

ID	User Story
US-01	As a cattle/dairy operator, I want to record morning milk production so that my daily production record is maintained digitally.
US-02	As a cattle/dairy operator, I want to record evening milk production so that my daily production record is complete.
US-03	As a cattle/dairy operator, I want to save a production entry and receive confirmation so that I know the record was stored.
US-04	As a cattle/dairy operator, I want to view my monthly total milk production so that I can quickly understand overall output.
US-05	As a cattle/dairy operator, I want to view my monthly average milk production so that I can understand typical production performance.
US-06	As a cattle/dairy operator, I want to compare morning and evening production so that I can understand production patterns.
US-07	As a cattle/dairy operator, I want production analytics to be easy to read so that I can understand the information without confusion.
US-08	As a cattle/dairy operator in an area with unreliable internet, I want core production functions to work offline so that I can maintain records consistently.

13. Acceptance Criteria — Initial

Story / Requirement	Acceptance Criteria
US-01 / FR-01	A user can select the relevant cattle/date and enter a morning production value; invalid or empty input is handled clearly; the value can be saved.
US-02 / FR-02	A user can enter an evening production value for the relevant cattle/date; invalid input is handled clearly; the value can be saved.
US-03 / FR-03	After saving a valid record, the product provides clear confirmation and the record appears in the relevant history.
US-04 / FR-05	For a selected month, the product displays a total based on the saved production records for that period.
US-05 / FR-06	For a selected month with valid records, the product displays an average based on the saved production records.

US-06 / FR-07	The product presents morning and evening production in a form that allows the user to compare them without additional manual calculation.
US-07 / FR-08	Primary production values and labels are readable; chart/context elements clearly communicate what is being measured.
US-08 / FR-09	Core production entry and review can be completed without an active internet connection.

14. Product Flow — Proposed V2

1. Open GokulaHealth.
2. Select or identify the cattle record.
3. Enter morning and/or evening milk production.
4. Save the record and receive confirmation.
5. Open production summary.
6. Select the desired month.
7. Review monthly total and monthly average.
8. Compare morning vs evening production.
9. Use clearly labeled analytics to understand performance.
10. Provide feedback or identify an unclear insight for later iteration.

15. Success Metrics / KPIs

The following are proposed metrics to validate whether V2 is improving the product. Baselines should be established before claiming improvement.

Metric	What It Measures	Measurement Direction
Production entry completion	Percentage of initiated production-entry tasks successfully completed.	Baseline + improvement after V2 testing.
Monthly summary comprehension	Percentage of test users who can correctly identify monthly total and average.	Target to be established during usability testing.
Time to understand monthly performance	Time required for a user to answer a defined monthly-production question.	Baseline V1 vs V2 comparison.
Record accuracy	Percentage of test entries correctly reflected in saved records and summaries.	Target should be defined before release.
Analytics usability	User-reported ease of understanding the main production information.	Baseline V1 vs V2 comparison.
Offline workflow success	Percentage of core production tasks completed without network access.	Target should be defined through testing.

16. Prioritization Principle

> ***Prioritize improvements that directly reduce friction in production recording or improve understanding of monthly production performance.***

The first prioritization pass should favor the core production workflow, monthly total/average, morning-versus-evening comparison, and analytics clarity. Lower-confidence features such as advanced analytics, cattle-level trends, and factory comparison should require stronger evidence before receiving high priority.

17. Dependencies & Risks

- Additional user research with active cattle/dairy operators is needed to validate the target workflow.
- Usability testing is needed to establish whether the proposed summaries are actually easier to understand.
- Production calculations depend on consistent and correctly stored underlying records.
- Offline behavior must be tested across realistic data-entry scenarios.
- Feature priorities may change if additional research contradicts the current assumptions.
- Because the current research sample is small, the PRD should remain adaptable rather than treating every requirement as permanently fixed.

18. Release / Validation Approach

1. Prototype the proposed V2 production workflow.
2. Test the prototype with representative users where possible.
3. Review usability findings and revise requirements.
4. Implement the highest-priority validated requirements.
5. Run functional and usability testing.
6. Release the focused V2 improvement.
7. Collect feedback and metrics.
8. Use evidence to decide whether a V2.1 iteration is justified.

19. V2.1 Iteration Candidates

- Validated improvements to cattle-level trends.
- Validated factory-level comparison.
- Additional production insights based on user feedback.
- Further usability improvements.
- New capabilities supported by evidence from V2 usage.

20. Traceability to Discovery

Discovery Evidence / Observation	V2 Requirement Direction	Artifact Link
Monthly calculation/analysis may take time.	Automated monthly total and average.	User Research → FR-05/FR-06
Automatic monthly calculation/analysis may be useful.	Provide clear monthly summaries.	User Research → Product Goal
Morning/evening production identified as priority information.	Provide morning vs evening comparison.	User Research → FR-07
V1 analytics can be difficult to understand.	Improve readability, labels, context.	V1 Analysis → FR-08
Cattle-level trend is unclear.	Validate before prioritizing redesign.	V1 Analysis → Secondary Scope
Factory comparison is incomplete/unclear.	Validate before further investment.	V1 Analysis → Secondary Scope

21. Assumptions Requiring Validation

- Manual production recording is a meaningful problem for active target users.
- Monthly total and average are among the most valuable production summaries.
- Users will consistently record morning and evening production.
- Clearer analytics will improve understanding of production performance.
- Offline operation is important in the target usage environment.
- The proposed workflow is suitable for the primary target segment.

22. Product Decision

> ***V2 should focus on making the core milk-production workflow easier to use and making monthly production performance easier to understand, while keeping lower-confidence analytics and comparison features behind validation.***

This is an iterative V2 definition based on the current evidence. Additional research and usability testing may change the requirements before implementation.

23. Document Status & Next Step

Status: Draft for validation.

Next PM artifacts: Prioritization and Roadmap will translate the PRD requirements into an ordered V2 delivery plan, followed by Figma design and Jira/Scrum execution.

The PRD should be updated if new research changes the target user, problem definition, priorities, or success criteria.

Document Control

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